observed anti-inflammatory effects of antagonists to these molecules, thus highlighting the importance of basic biology to the practice of medicine. The redundancy of the mediators and their actions ensures that this protective response remains robust and is not readily subverted.

SUMMARY

ACTIONS OF THE PRINCIPAL MEDIATORS OF INFLAMMATION

- Vasoactive amines, mainly histamine: vasodilation and increased vascular permeability
- Arachidonic acid metabolites (prostaglandins and leukotrienes): several forms exist and are involved in vascular reactions, leukocyte chemotaxis, and other reactions of inflammation; antagonized by lipoxins
- Cytokines: proteins produced by many cell types; usually act at short range; mediate multiple effects, mainly in leukocyte recruitment and migration; principal ones in acute inflammation are TNF, IL-1, and chemokines
- Complement proteins: Activation of the complement system by microbes or antibodies leads to the generation of multiple breakdown products, which are responsible for leukocyte chemotaxis, opsonization and phagocytosis of microbes and other particles, and cell killing
- Kinins: produced by proteolytic cleavage of precursors; mediate vascular reaction, pain

MORPHOLOGIC PATTERNS OF ACUTE INFLAMMATION

The morphologic hallmarks of acute inflammatory reactions are dilation of small blood vessels and accumulation of leukocytes and fluid in the extravascular tissue. The vascular and cellular reactions account for the signs and symptoms of the inflammatory response. Increased blood flow to the injured area and increased vascular permeability lead to the accumulation of extravascular fluid rich in plasma proteins (*edema*) and account for the redness *(rubor)*, warmth *(calor)*, and swelling *(tumor)* that accompany acute inflammation. Leukocytes that are recruited and activated by the offending agent and by endogenous mediators may release toxic metabolites and proteases extracellularly, causing tissue damage and loss of function (*functio laesa*). During the damage, and in part as a result of the liberation of prostaglandins, neuropeptides, and cytokines, one of the local symptoms is pain *(dolor).*

Although these general features are characteristic of most acute inflammatory reactions, special morphologic patterns are often superimposed on them, depending on the severity of the reaction, its specific cause, and the particular tissue and site involved. The importance of recognizing distinct gross and microscopic patterns of inflammation is that they often provide valuable clues about the underlying cause.

Serous Inflammation

Serous inflammation is marked by the exudation of cell-poor fluid into spaces created by injury to surface epithelia or into body cavities lined by the peritoneum, pleura, or pericardium. Typically, the fluid in serous inflammation is not infected by destructive organisms and does not contain large numbers of leukocytes (which tend to produce purulent inflammation, described later). In body cavities the fluid may be derived from the plasma (as a result of increased vascular permeability) or from the secretions of mesothelial cells (as a result of local irritation); accumulation of fluid in these cavities is called an *effusion*. (Effusions consisting of transudates also occur in noninflammatory conditions, such as reduced blood outflow in heart failure, or reduced plasma protein levels in some kidney and liver diseases.) The skin blister resulting from a burn or viral infection represents accumulation of serous fluid within or immediately beneath the damaged epidermis of the skin (Fig. 3.12).

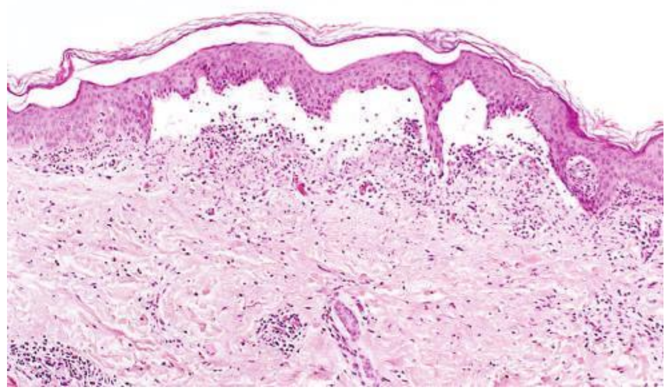

Fig. 3.12 Serous inflammation. Low-power view of a cross section of a skin blister showing the epidermis separated from the dermis by a focal collection of serous effusion.

Fibrinous Inflammation

A fibrinous exudate develops when the vascular leaks are large or there is a local procoagulant stimulus. With a large increase in vascular permeability, higher-molecular-weight proteins such as fibrinogen pass out of the blood, and fibrin is formed and deposited in the extracellular space. A fibrinous exudate is characteristic of inflammation in the lining of body cavities, such as the meninges, pericardium (Fig. 3.13A), and pleura. Histologically, fibrin appears as an eosinophilic meshwork of threads or sometimes as an amorphous coagulum (Fig. 3.13B). Fibrinous exudates may be dissolved by fibrinolysis and cleared by macrophages. If the fibrin is not removed, with time, it may stimulate the ingrowth of fibroblasts and blood vessels and thus lead to scarring. Conversion of the fibrinous exudate to scar tissue *(organization)* within the pericardial sac leads to opaque fibrous thickening of the pericardium and epicardium in the area of exudation and, if the fibrosis is extensive, obliteration of the pericardial space.

Purulent (Suppurative) Inflammation, Abscess

Purulent inflammation is characterized by the production of pus, an exudate consisting of neutrophils, the liquefied debris of necrotic cells, and edema fluid. The most frequent cause of purulent (also called *suppurative*)

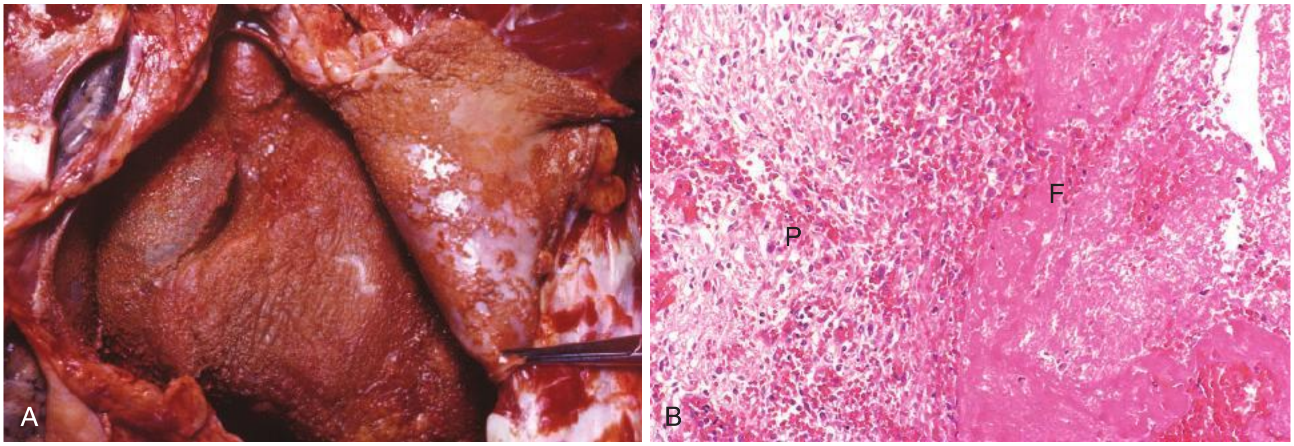

Fig. 3.13 Fibrinous pericarditis. (A) Deposits of fibrin on the pericardium. (B) A pink meshwork of fibrin exudate *(F)* overlies the pericardial surface *(P)*.

inflammation is infection with bacteria that cause liquefactive tissue necrosis, such as staphylococci; these pathogens are referred to as *pyogenic* (pus-producing) bacteria. A common example of an acute suppurative inflammation is acute appendicitis. ***Abscesses* are localized collections of pus** caused by suppuration buried in a tissue, an organ, or a confined space. They are produced by seeding of pyogenic bacteria into a tissue (Fig. 3.14). Abscesses have a central region that appears as a mass of necrotic leukocytes and tissue cells. There is usually a zone of preserved neutrophils around this necrotic focus, and outside this region there may be vascular dilation and parenchymal and fibroblastic proliferation, indicating chronic inflammation and repair. In time the abscess may become walled off and ultimately replaced by connective tissue. When persistent or at critical locations (such as the brain), abscesses may have to be drained surgically.

Ulcers

An ulcer is a local defect, or excavation, of the surface of an organ or tissue that is produced by the sloughing (shedding) of inflamed necrotic tissue (Fig. 3.15). Ulceration can occur only when tissue necrosis and resultant inflammation exist on or near a surface. It is most commonly encountered in (1) the mucosa of the mouth, stomach, intestines, or genitourinary tract, and (2) the skin and subcutaneous tissue of the lower extremities in older persons who have circulatory disturbances that predispose to extensive ischemic necrosis. Acute and chronic inflammation often coexist in ulcers, such as peptic ulcers of the stomach or duodenum and diabetic ulcers of the legs. During the acute stage there is intense polymorphonuclear infiltration and vascular dilation in the margins of the defect. With chronicity, the margins and base of the ulcer develop fibroblast proliferation, scarring, and the accumulation of lymphocytes, macrophages, and plasma cells.

OUTCOMES OF ACUTE INFLAMMATION

Although, as might be expected, many variables may modify the basic process of inflammation, including the

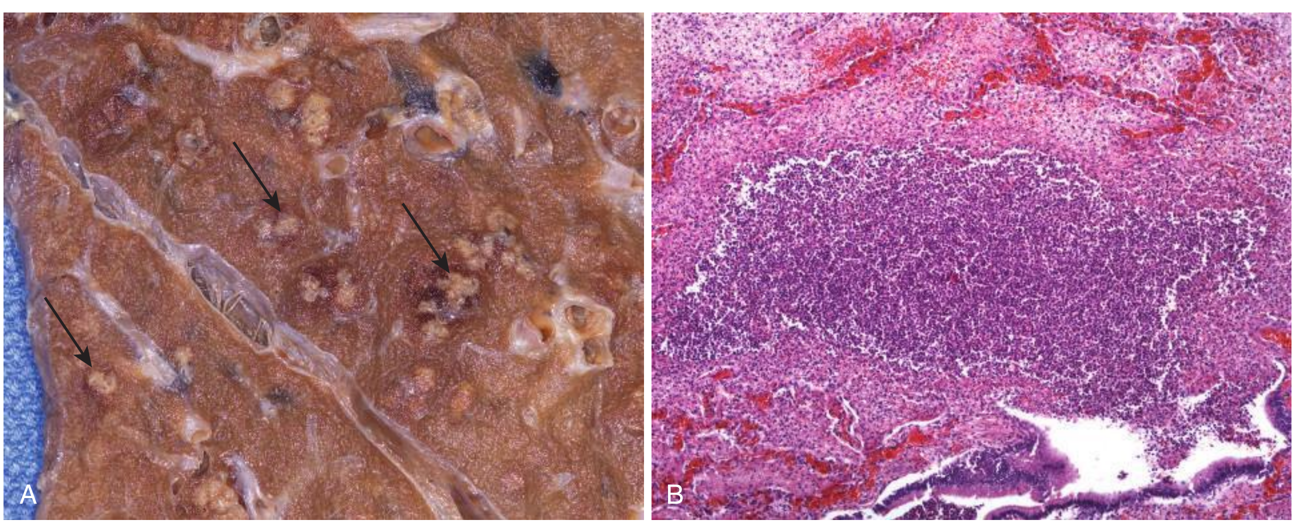

Fig. 3.14 Purulent inflammation. (A) Multiple bacterial abscesses *(arrows)* in the lung in a case of bronchopneumonia. (B) The abscess contains neutrophils and cellular debris, and is surrounded by congested blood vessels.

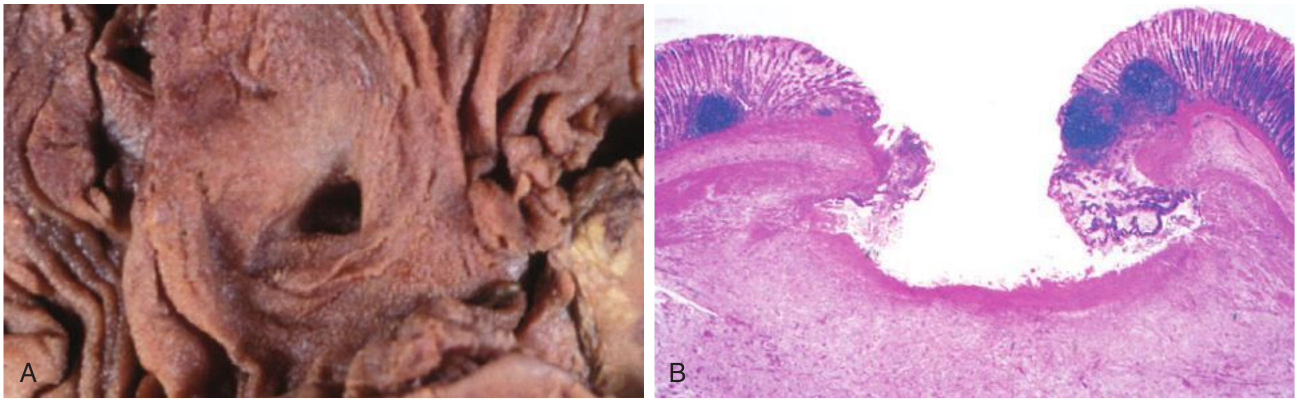

Fig. 3.15 The morphology of an ulcer. (A) A chronic duodenal ulcer. (B) Low-power cross-section view of a duodenal ulcer crater with an acute inflammatory exudate in the base.

nature and intensity of the injury, the site and tissue affected, and the responsiveness of the host, **acute inflammatory reactions typically have one of three outcomes** (Fig. 3.16):

- **Complete resolution.** In a perfect world, all inflammatory reactions, after they have succeeded in eliminating the offending agent, should end with restoration of the site of acute inflammation to normal. This is called *resolution* and is the usual outcome when the injury is limited or short-lived or when there has been little tissue destruction and the damaged parenchymal cells can regenerate. Resolution involves removal of cellular debris and microbes by macrophages, and resorption of edema fluid by lymphatics.
- **Healing by connective tissue replacement (scarring,** or **fibrosis).** This occurs after substantial tissue destruction, when the inflammatory injury involves tissues that are incapable of regeneration, or when there is abundant fibrin exudation in tissue or in serous cavities (pleura, peritoneum) that cannot be adequately cleared. In all

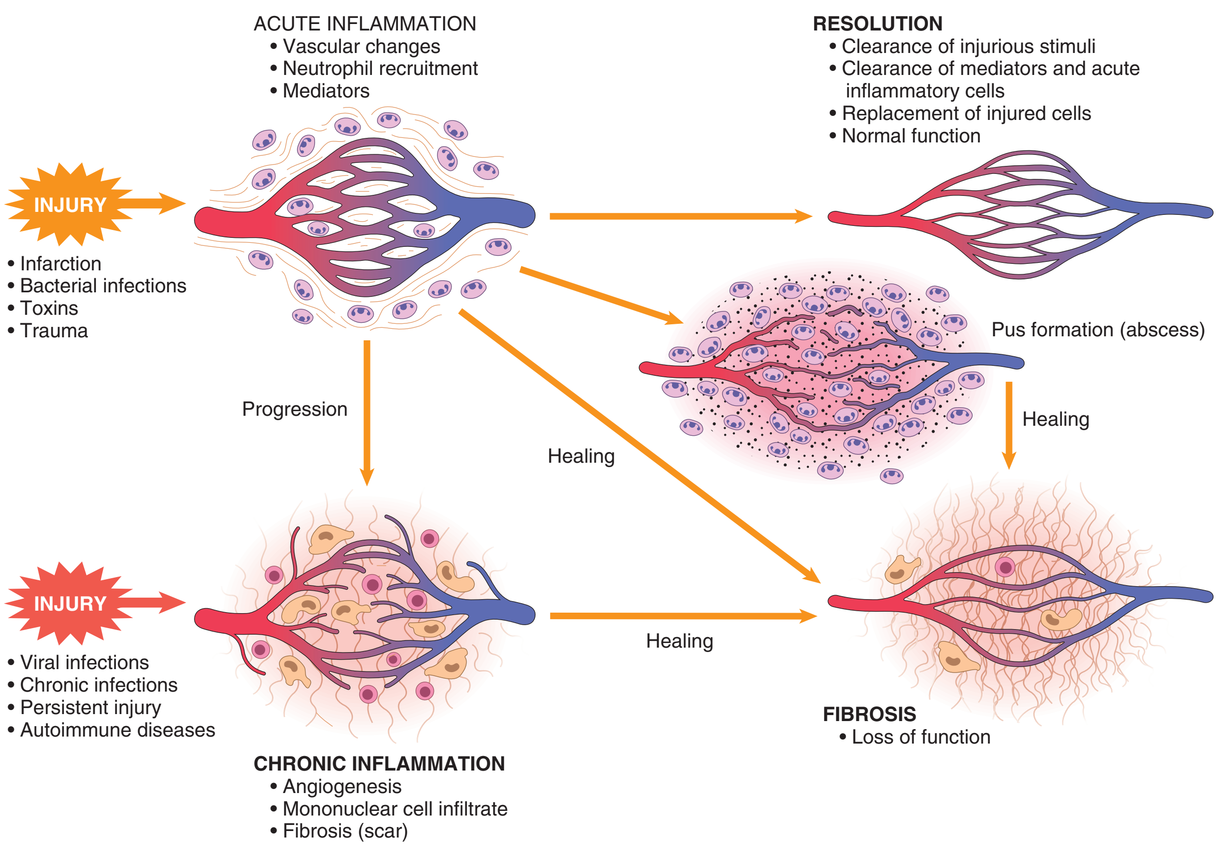

Fig. 3.16 Outcomes of acute inflammation: resolution, healing by fibrosis, or chronic inflammation. The components of the various reactions and their functional outcomes are listed.

these situations, connective tissue grows into the area of damage or exudate, converting it into a mass of fibrous tissue.

- Progression of the response to **chronic inflammation.** Acute to chronic transition occurs when the acute inflammatory response cannot be resolved, as a result of either the persistence of the injurious agent or some interference with the normal process of healing.

CHRONIC INFLAMMATION

Chronic inflammation is a response of prolonged duration (weeks or months) in which inflammation, tissue injury, and attempts at repair coexist, in varying combinations. It may follow acute inflammation, as described earlier, or may begin insidiously, as a smoldering, sometimes progressive, process without any signs of a preceding acute reaction.

Causes of Chronic Inflammation

Chronic inflammation arises in the following settings:

- **Persistent infections** by microorganisms that are difficult to eradicate, such as mycobacteria and certain viruses, fungi, and parasites. These organisms often evoke an immune reaction called *delayed-type hypersensitivity* (Chapter 5). The inflammatory response sometimes takes a specific pattern called *granulomatous inflammation* (discussed later). In other cases, unresolved acute inflammation evolves into chronic inflammation, such as when an acute bacterial infection of the lung progresses to a chronic lung abscess.
- **Hypersensitivity diseases.** Chronic inflammation plays an important role in a group of diseases that are caused by excessive and inappropriate activation of the immune system (Chapter 5). In *autoimmune diseases*, self (auto) antigens evoke a self-perpetuating immune reaction that results in chronic inflammation and tissue damage; examples of such diseases are rheumatoid arthritis and multiple sclerosis. In *allergic diseases*, chronic inflammation is the result of excessive immune responses against common environmental substances, as in bronchial asthma. Because these autoimmune and allergic reactions are triggered against antigens that are normally harmless, the reactions serve no useful purpose and only cause disease. Such diseases may show morphologic patterns of mixed acute and chronic inflammation because they are characterized by repeated bouts of inflammation. Fibrosis may dominate the late stages.
- **Prolonged exposure to potentially toxic agents, either exogenous or endogenous.** An example of an exogenous agent is particulate silica, a nondegradable inanimate material that, when inhaled for prolonged periods, results in an inflammatory lung disease called *silicosis* (Chapter 13). *Atherosclerosis* (Chapter 10) is a chronic inflammatory process affecting the arterial wall that is thought to be induced, at least in part, by excessive production and tissue deposition of endogenous cholesterol and other lipids.
- Some forms of chronic inflammation may be important in the pathogenesis of diseases that are not conventionally thought of as inflammatory disorders. These include neurodegenerative diseases such as Alzheimer disease, metabolic syndrome and the associated type 2 diabetes, and certain cancers in which inflammatory reactions promote tumor development. The role of inflammation in these conditions is discussed in the relevant chapters.

Morphologic Features

In contrast to acute inflammation, which is manifested by vascular changes, edema, and predominantly neutrophilic infiltration, chronic inflammation is characterized by the following:

- **Infiltration with mononuclear cells,** which include macrophages, lymphocytes, and plasma cells (Fig. 3.17)
- **Tissue destruction,** induced by the persistent offending agent or by the inflammatory cells
- **Attempts at healing** by connective tissue replacement of damaged tissue, accomplished by *angiogenesis* (proliferation of small blood vessels) and, in particular, *fibrosis*

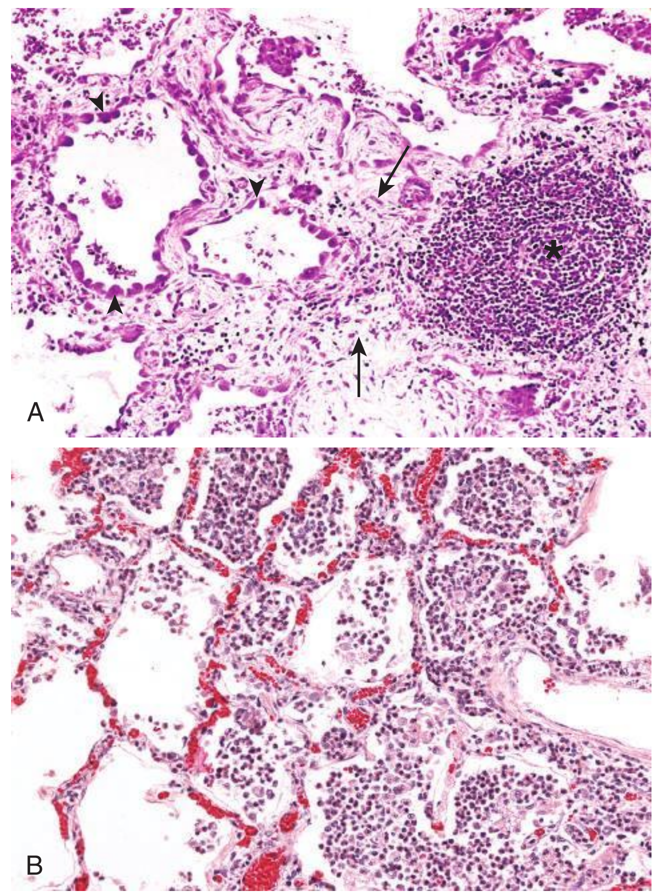

Fig. 3.17 (A) Chronic inflammation in the lung, showing all three characteristic histologic features: (1) collection of chronic inflammatory cells (***), (2) destruction of parenchyma (normal alveoli are replaced by spaces lined by cuboidal epithelium, *arrowheads*), and (3) replacement by connective tissue (fibrosis, *arrows*). (B) In contrast, in acute inflammation of the lung (acute bronchopneumonia), neutrophils fill the alveolar spaces and blood vessels are congested.